Roll Number

JNTUA COLLEGE OF ENGINEERING (Autonomous) PULIVENDULA

Department Name: CIVIL, EEE, ME, ECE & CSE

LINEAR ALGEBRA AND CALCULUS

Descriptive Paper for II[nd] Mid Examination, January 2024

Date: 18-01-2024

Marks: 30

Duration: 90 Min

Answer the following Either / Or Questions Each carries 10 Marks

		UNIT – III	
1.	a)	Verify Rolle's theorem for $f(x)=(x-a)^m(x-b)^n$, in $[a,b]$, $m>1, n>1$.	[5 M]
	b)	Use Lagrange mean value theorem to prove that if $0<a<b<1$, $\frac{b-a}{1+b^2}<\tan^{-1}b-\tan^{-1}a<\frac{b-a}{1+a^2}$. Hence show that $\frac{\pi}{4}+\frac{3}{25}<\tan^{-1}\frac{4}{3}<\frac{\pi}{4}+\frac{1}{6}$.	[5 M]
		OR	
2.	a)	Verify Cauchy's Mean-value theorem for the functions $f(x)=e^x$ and $g(x)=e^{-x}$ in the interval (a,b).	[5 M]
	b)	Using the Maclaurin series, expand $f(x)=\tan x$ up to the term containing x^5.	[5 M]
		UNIT – IV	
3.	a)	If $u=x^2+y^2+z^2$ and $x=e^t$; $y=\sin t$; $z=\log t$, Find $\frac{du}{dt}$ using chain rule and verify the result by direct substitution.	[5 M]
	b)	Find the maximum and minimum values of $f(x,y)=x^4+y^4-2x^2+4xy-2y^2$.	[5 M]
		OR	
4.	a)	Expand $f(x,y)=e^y\log(1+x)$ in powers of x and y.	[5 M]
	b)	If $u=\frac{x+y}{1-xy}$ and $v=\tan^{-1}x+\tan^{-1}y$, find $\frac{\partial(u,v)}{\partial(x,y)}$. Are u and v functionally related? If so, find this relationship.	[5M]
		UNIT – V	
5.	a)	By changing the order of integration, and hence evaluate $\int_0^1\int_{x^2}^{2-x} xy\,dy\,dx$.	[5 M]
	b)	Evaluate $\int_{-1}^{1}\int_0^z\int_{x-z}^{x+z}(x+y+z)\,dy\,dx\,dz$.	[5 M]
		OR	
6.	a)	By changing into polar coordinates, evaluate $\int_0^\infty\int_0^\infty e^{-(x^2+y^2)}dx\,dy$.	[5 M]
	b)	Using Triple Integral, Find the volume of the sphere $x^2+y^2+z^2=a^2$.	[5 M]

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JNTUA COLLEGE OF ENGINEERING: PULIVENDULA (AUTONOMOUS)

I.B.Tech. I- SEM. II MID TEST, Jan-2024

Subject: COMMUNICATIVE ENGLISH

Time: 90 Min. **BRANCHES (,EEE&CSE)** **Max. Marks.15**

Answer the following questions selecting *one* from each unit. Each question carries *five* marks: **Marks: 3x5=15**

UNIT-III

1.(a) Elon Musk's life story is a compelling source of inspiration for many people. Explain?

(b) What are the techniques of paraphrasing?

OR

2.(a) What is the significance of SpaceX and Tesla in Elon Musk'story?

(b) What are the features of a good summary?

UNIT-IV

3.(a) What is the central idea of the lesson 'The Toys of Peace'.

(b) Differentiate intensive reading and extensive reading.

OR

4.(a) What is skimming and scanning?

(b) 'Actions speak louder than words'. Explain in your own words?

UNIT-V

5.(a) Describe the structure of an official letter?

(b) Write an imaginary conversation between two friends about upcoming elections?

OR

6.(a) What are some advantages of intrapersonal communication?

(b) What are the best ways to learn more about body language?

JNTUA COLLEGE OF ENGINEERING (AUTONOMOUS) :: PULIVENDULA

I B. Tech, I Semester II – INTERNAL TEST – JANUARY - 2024

INTRODUCTION TO PROGRAMMING

(CSE)

Time: 90 Min. **Marks: 15M**

Answer all the following Questions.

1. A) What is an array? Explain 2D arrays in detail. 3M

   B) Write a C program to find sum, average and count of odd numbers in a given matrix. 2M

OR

2. A) What is a string? Explain the memory model of 2D array with an example. 3M

   B) Write a C program to perform concatenation of two strings without using built-in function. 2M

3. A) What is a pointer? Write a C program to find length of a string using functions. 3M

   B) What is user-defined data type? Differentiate structures and unions with a suitable example. 2M

OR

4. A) What is a structure? Explain the concept of self-referential structures. 3M

   B) Explain about pointer and address arithmetic with an example. 2M

5. What is a function? Discuss about different types of function calls with examples. 5M

OR

6. A) What is a file? Explain file I/O operations. 3M

   B) Write a C program to find sum of diagonal elements of a matrix using functions. 2M

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(Common to EEE & CSE)

DATE: 19.01.2024 (FN)

TIME: 90Mts. Descriptive Max. Marks :30M

Answer the following questions

All questions carries equal marks

Part – A

1. a) Define Levelling and Explain the Different types of levelling 5M

b) Define Bench Mark, back sight and List out various types of Bench marks. 5M

(OR)

2.a) Define Transportation Engineering and Explain the Different types of pavements in Highways with neat sketches 10M

Part – B

3. a) With a suitable P-V & T-S diagram give a detail Comparission between Otto cycle & Diesel Cycle 5M

b) Give a different detail mechanical power transmission systems and list out various applications 5M

(OR)

4. a) With a suitable sketch explain the working principle of steam power plant. 5M

b) With a suitable sketch explain different types of boilers. Explain their working principle. 5M

Part – A & Part – B

5. a) Explain the sources of water and write the importance of water in Daily Life. 5M

b) Define Rainwater Harvesting and Write 4 Differences between Dam and Reservoir. 5M

(OR)

6. a) With a suitable sketch explain the working principle of nuclear power plant 5M

b) Explain various components of Electric and Hybrid Vehicles 5M

Note:

Kindly write your answer for **Part A & Part B** Separately

23191F

JNTUA COLLEGE OF ENGINEERING (Autonomous) PULIVENDULA

Department Name: Humanities and Basic sciences

Chemistry – (23ABS03)

Descriptive Paper for II - Mid Examination

Class: I.B.Tech **Marks: 30**

Branch: E.E.E &C.S.E **Date:18-01-2024(A.N)** **Duration: 90 Min**

Answer the following Either / Or Questions. Each question carries 10 Marks

UNIT – III

1. Define conductivity. Write a detail note on Conduct metric titration.

(Or)

2. Write a brief notes on i) Lithium ion battery. ii) H_2-O_2 Fuel cell

UNIT – IV

3. Explain the chain growth polymerization mechanism.

(Or)

4. Write preparation, properties and applications of Teflon and Buna-N.

UNIT – V

5. Write a detail note on instrumentation and applications of Uv- Visible spectroscopy.

(Or)

6. What is chromatography? Write instrumentation and applications of HPLC.

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